

Sihan Qin

[Email](#) | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#) | [Homepage](#) | Guangzhou, P.R. China

EDUCATION

Sun Yat-sen University

Guangzhou, China

B.E. in Computer Science and Technology (Artificial Intelligence & Big Data)

Jun. 2027(Expected)

GPA: 3.8/4.0

- **Core Courses:** Machine Learning & Data Mining, Introduction to Robotics, Modern Control Systems, Principles & Applications of Multimodal Deep Models, Artificial Intelligence, Optimization Theory, Pattern Recognition
- **Awards:** 3rd-Class University Scholarship (2023, 2024); Honorable Mention, Mathematical Contest in Modeling 2025

RESEARCH EXPERIENCE

RL Post-Training for VLA Models in Multi-task Settings

Shenzhen, China

[ISEE Lab](#), Sun Yat-sen University, advised by Prof. [Weishi Zheng](#)

Apr. 2026 – Sep. 2026

- Trained and evaluated VLA models (StarVLA & $\pi_{0.5}$) with RL Token in simulated environments, including RoboTwin2.0 and LIBERO.
- Building a real-world end-to-end $\pi_{0.5}$ (with RL Token) pipeline for AgileX NERO/Piper arms, spanning model deployment, teleoperated data collection, training, and online inference.

[Physical Autoregressive Model for Robotic Manipulation without Action Pretraining](#)

Guangzhou, China

[HCP Lab](#), Sun Yat-sen University, advised by Prof. [Guangrun Wang](#) and Prof. [Liang Lin](#)

Jul. 2025 - Mar. 2026

- Proposed and developed a physical autoregressive model that transfers physical priors from video pretraining to robotic manipulation, enabling video-consistent action generation without action pretraining.
- Conducted training and evaluation on ManiSkill, LIBERO, and CALVIN; tuned key hyperparameters including batch size, learning rate, and LR scheduler, improving simulation success rates from about 30% to over 90%.
- Conducted real-robot experiments on a Franka Panda arm and collected 80 teleoperated trajectories for supervised learning, improving model generalization and real-world robustness; also contributed to paper writing.

Procedural 3D Image Generation from Lung CT Scans

Guangzhou, China

[Robotic and Intelligence Computing Lab](#), Sun Yat-sen University, advised by Prof. [Kai Huang](#)

May - Sep. 2024

- Curated 20 full respiratory-cycle lung CT samples from public datasets and performed threshold segmentation, meshing, and point cloud conversion with scikit-image and Open3D.
- Separated bronchi and lung boundaries using RANSAC, curvature clustering, Alpha Shape, and voxel filtering to support downstream lesion analysis.

PUBLICATION

Zijian Song, **Sihan Qin**, Tianshui Chen, Liang Lin, Guangrun Wang, *Physical Autoregressive Model for Robotic Manipulation without Action Pretraining*, [arXiv:2508.09822](#).

Zijian Song, Qichang Li, **Sihan Qin**, Yuhao Chen, Tianshui Chen, Liang Lin, Guangrun Wang, *Learning Physics from*

Pretrained Video Models: A Multimodal Continuous and Sequential World Interaction Models for Robotic Manipulation, [arXiv:2603.00110](https://arxiv.org/abs/2603.00110), **icmr 2026**.

Zijian Song, Xiaoxin Lin, Qiuming Huang, **Sihan Qin**, Guangrun Wang, Liang Lin, *SIRI-Bench: Challenging VLMs' Spatial Intelligence through Complex Reasoning Tasks*, [arXiv:2506.14512](https://arxiv.org/abs/2506.14512).

INTERNSHIP EXPERIENCE

BrainCo Inc Shenzhen, China
Embodied AI Algorithm Intern Sep.2026 – present

- Built an end-to-end data collection platform for robotic prosthetics within the first three weeks, integrating a MANUS teleoperation glove, a BrainCo Revo3 dexterous hand, a D405 camera, and an electromyography (EMG) armband.
- Implemented hybrid motion retargeting with GeoRT for the thumb and AnyTeleop for the other four fingers, and established time synchronization across teleoperation, visual, and EMG data streams.
- Supporting a vision-language-action (VLA) + EMG prosthetics project aimed at preserving user control through EMG-based action selection and enabling dexterous manipulation across objects and scenes through VLA generalization.

Guangzhou Shuzhi Technology Guangzhou, China
Algorithm Intern Sep. - Oct. 2025

- Contributed to prompt engineering for an academic-industry matching platform.
- Diagnosed key flaws in legacy AI interactions and implemented a prompt framework combining UltraRAG + Deep Research with multi-round retrieval-LLM for deep analysis of project value, team structure, and research outcomes.
- Structured 5 key LLM-generated analytical results into JSON and stored them in databases to support front-end development.

OTHERS

Zotero Paper Partner Guangzhou, China
Independent Project Apr. 2026

- Developed an attention-friendly Zotero plugin that answers questions inside native notes.

Mathematical Contest in Modeling 2025 Guangzhou, China
Olympic Medal Table Prediction, Honorable Mention Feb. 2025

- Built regression, random forest, and SVM models, combined with SMOTE, Bayesian optimization, and Monte Carlo simulation, to predict medal counts and analyze key influencing factors.

CAMPUS LEADERSHIP

Sun Yat-sen University Student Union & School of Computer Science Student Union Sep. 2023 – Jul. 2024
General Affairs Department Member; "Red Kapok" Event Coordinator

SKILLS

Technical: Machine Learning, Deep Learning, Robotics, PyTorch, Linux

- **Real Robot:** Franka, Agile-X(Nero/Piper) and so-101
- **Simulation Platform:** ManiSkill, LIBERO and RoboTwin2.0

- **VLA model:** $\pi_{0.5}$, StarVLA, smolVLA

Languages: Mandarin (Native), Cantonese (Fluent), English (Fluent), French (Basic)